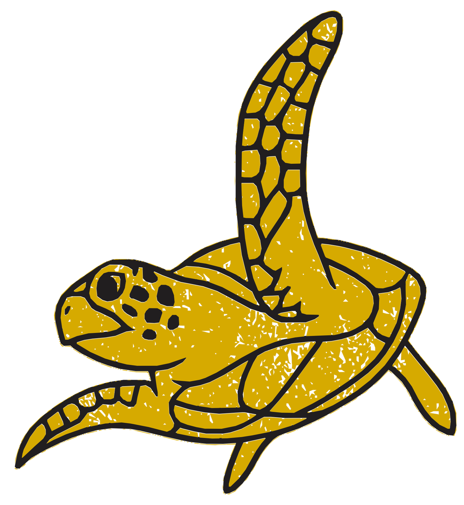


Hypersperical Fitness Functions

for Many-Objective Optimisation

As the number of objectives grows too large, dominance collapses.

What happens if we abandon dominance - and turn to hyperspherical geometry?

We propose casting vectors of fitness values onto a hypersphere, using an L2 Norm, and measuring the angle between the genetic individual's fitness point and the True North objective, which we define as:

Augmented projection: append a 1, then L2-normalise — a lift into m+1 dimensions

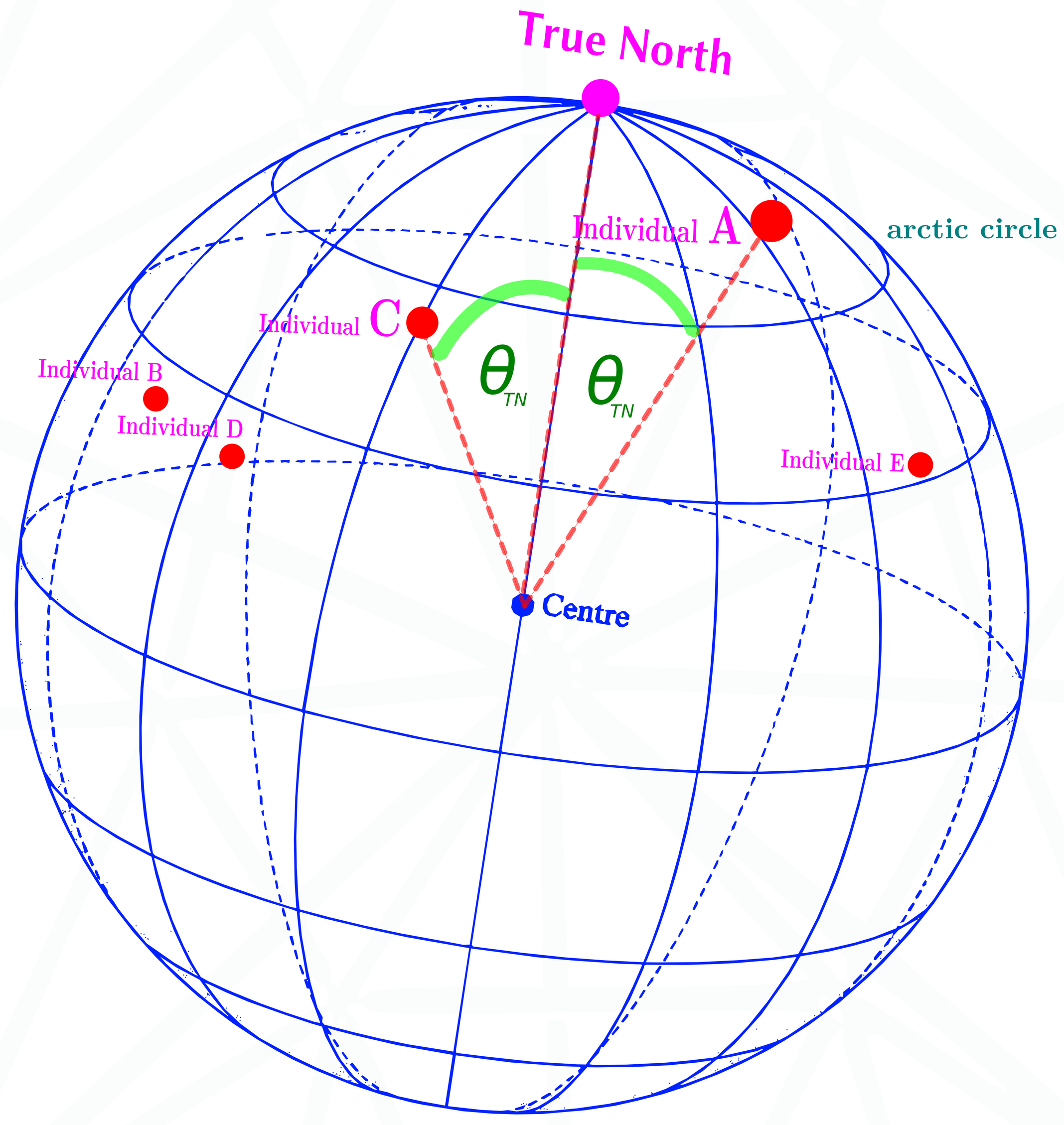
$$\mathbf{y} = \frac{\mathbf{x}}{\|\mathbf{x}\|}, \quad e = \max\left(0,\ 1 - \frac{\|\mathbf{x}\|^2}{m}\right), \quad \mathbf{y}' = \left(\sqrt{1-e^2}\ \mathbf{y},\ e\right) \in S^m$$

$$\mathbf{p}_{TN} = (0,\ \ldots,0,1), \quad \theta_{TN} = \arccos(\mathbf{y}' \cdot \mathbf{p}_{TN}) = \arccos(e)$$

$e \in [0,1]$ is the energy score: $e = 1$ at True North (ideal, $\|\mathbf{x}\| = 0$); $e \to 0$ as energy grows

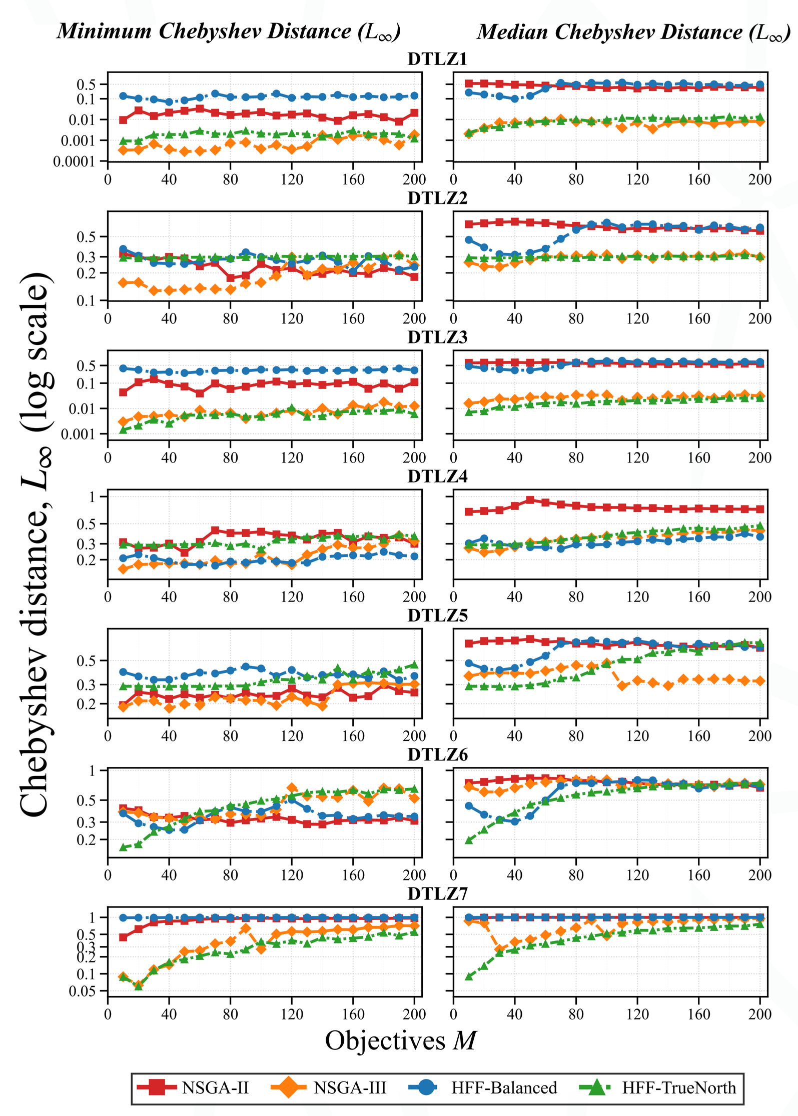
Doing so, we scalarise large numbers of objectives into a single angle. Our genetic population evolves toward optimal solutions, minimising this angle using single-objective GA. Support for n objectives.

Individual A wins!



We use cosine similarity to run tournaments across N Objectives

Does it work? It's hard to measure fairly.



We measured mean worst-case regret: Yu (1973). It's not favouring angular or dominance-based methods.

TrueNorth 0.317 vs NSGA-III: 0.355

Tighter spread (CoV 75.5% vs 80.6%).

Performance is a great reason to try HFF

Selection cost per generation

HFF — ours

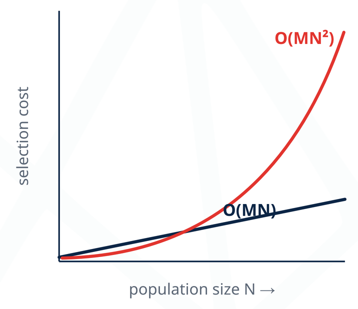
O(MN)

linear in population N and objectives M

NSGA-II / NSGA-III

O(MN²)

quadratic in N — non-dominated sort



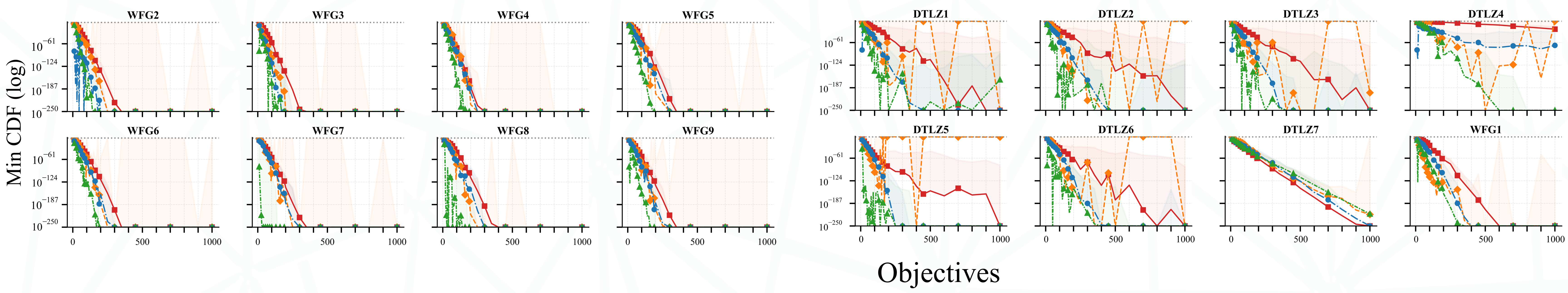
Measured: HFF-Balanced 144,842 s total vs NSGA-III 433,535 s — 3× slower, superlinear in N.

get the code!

We developed a hyperspherical measure, that corrects for the concentration of measure

WFG

DTLZ



Min CDF (log)

$P(\theta \le t) = I_{\sin^2 t}\left(\frac{m-1}{2}, \frac{1}{2}\right)$

github.com/gamakon/hff

ABSTRACT

Many-objective optimization with more than three objectives causes selection pressure collapse in Pareto-based algorithms: nearly all solutions become mutually non-dominated. We introduce hyperspherical fitness functions (HFF), angular scalarization methods that project objective vectors onto the unit hypersphere and rank solutions by angular distance to a target direction. Two variants are presented: HFF-TrueNorth targets minimization via an augmented projection encoding both direction and magnitude, while HFF-Balanced targets equitable trade-offs via a balanced pole. Both achieve O(MN) selection complexity versus O(MN²) for NSGA-II and NSGA-III selection. On DTLZ1–7 and WFG1–9 (10 to 1000 objectives, 31 runs), HFF-TrueNorth outperforms NSGA-II and NSGA-III on CDF-corrected angular distance across the majority of problem-objective pairs. HFF targets operational many-objective problems — autonomous decision loops requiring a single actionable solution per cycle — where no human selects from a Pareto front. HFF-Balanced is the fastest algorithm tested, while NSGA-III is 3× slower and scales superlinearly. L∞ cross-validation confirms HFF-TrueNorth produces consistently low worst-case regret on a non-angular metric.

CITATION

Andrew James Morgan. 2026. Hyperspherical Fitness Functions for Many-Objective Optimization. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO Companion '26), July 13–17, 2026, San Jose, Costa Rica. ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3795101.3805445